DRAFT — UE26CY151A · Unit_03 · Energy storage devices, hydrogen energy, and sensors · Gas sensor - O2 sensor in automobile exhaust stream, Biosensors – Blood glucose sensor (Part 1)

Standalone video lesson · authored study notes with source citations

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· How to use this pack

Read the topic explanation, worked examples, and recap. Evidence citations point to source transcript intervals and slide pages; full slide images remain in the source visual index. Topics include the applicable parts: overview, key terms, formulas, figures, flowcharts, important points, worked examples, past-paper questions and a recap. IDs are shared with the separate documents: E = formula, G = graph, F = flowchart, W = worked example. Each ID is numbered by topic, so E4.2 is the second formula of topic 4.

Source text	Verbatim or OCR-derived text, kept separate from editorial notes.
Candidate	Automated topic/source association requiring manual confirmation.
Low ASR confidence	Transcript segment requires audio verification.
DRAFT	Human review remains incomplete.

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M1

Gas sensor - O2 sensor in automobile exhaust stream, Biosensors – Blood glucose sensor (Part 1)

1 Gas sensor - O2 sensor in automobile exhaust stream, Biosensors – Blood glucose sensor (Part 1)

Study focus: Gas sensor - O2 sensor in automobile exhaust stream, Biosensors – Blood glucose sensor (Part 1). This topic explains the central idea, its conditions, and how the lecture applies it in context. Review the selected source statements and follow their citations for the full lesson segment. EV0000170 · EV0000171

SHORT-NOTE RECAP

Core explanation · 00:00–04:59

We will continue with module five on energy storage devices and will discuss oxygen sensor.

Key details

- Firstly, we will understand what is an oxygen sensor and look at some applications of oxygen sensor.
- and specifically how is an oxygen sensor used in automobiles the principle construction and working.
- It is an electronic device that measures the proportion of oxygen in the gas or liquid being analyzed.
- First and foremost.
- Divers use it to measure the partial pressure of oxygen in their breathing gas.
- We know that in the atmosphere we have about twenty percent of oxygen, which is the right proportion for us to breathe.
- There in the atmosphere.
- Hence monitoring the partial pressure of oxygen is very important, and therefore they use oxygen sensors.
- They are equipped, the deep sea divers gas cylinders carrying oxygen are equipped with oxygen sensors.
- Another application is when scientists use as probes to measure respiration or production of oxygen.
- Oxygen analyzers used in.

- Many applications such as anesthesia monitors and respirators have to use these oxy sensors to determine the partial pressure of oxygen which is being used in these medical appliances.
- It is also used to measure the exhaust gas concentrations of oxygen in IC engines, that is in internal combustion engines.
- Mentioned here, that is to measure the exhaust gas concentration of oxygen in IC engines.
- When fuel burns the ideal ratio for combustion of air is to fuel would be fourteen point seven is to one.
- Why do we need this ratio?
- Because when combustion is complete, we know that the there will be only carbon dioxide and water which are going to be let out into the environment.
- into the atmosphere.
- But if the ratio is not at the stechometric air fuel ratio, then there is going to be other gases which will be let out into the atmosphere.
- atmosphere which are harmful to us.
- And therefore, if we want pollution to be minimum, then.
- Th it is the stechometric air fuel ratio which should be used in the IC engines.
- And of course this ratio is for gasoline.
- ratio so we know that we require the air to fuel ratio should be 14.7 is to one.
- Now the oxygen sensor therefore allows the engine control system to maintain this ideal ratio across various engine operating systems.
- We know that when an automobile is moving then.

Example or application from the lecture

- Let's define what is an oxygen sensor.
- So first let us understand the principle based on which the oxygen sensor works and specifically in automobiles.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000170 · EV0000171 · EV0000172 · +1 source units in pack.json

Core explanation · 05:00–09:59

The engine operating conditions keep changing based on whether it is moving slow, whether it is moving fast, whether it's going on a terrain which is not smooth.

Key details

- So every time the operating conditions are changing and the oxygen sensor therefore allows the engine control system to maintain the ideal ratio across all these conditions.
- It is going to compare the amount of oxygen in the exhaust to the amount of oxygen in the atmosphere.
- And we know that in the exhaust stream some amount of oxygen is going to be let out after the combustion reaction has happened.
- It is going to give an output voltage.
- Point two, then that represents a lean mixture.
- What do we mean by lean mixture?

- A lean mixture would be one which is lean in the fuel.
- That means that there is less of fuel and more of oxygen.
- Whereas if it is a rich mixture, rich mixture meaning it is rich in the fuel, there's lot of fuel and less of oxygen.
- The required amount of oxygen is not available and in that case the voltage is 0.8 volts.
- Now this is understandable because if it is a lean mixture that means there is more oxygen in the exhaust team.
- It is a rich mixture, there is going to be less oxygen in the exhaust stream, and when you compare it with the atmospheric oxygen, the difference is going to be higher.
- As the output voltage.
- The ideal set point would be around 0.45 volts.
- which it is receiving the signal voltage signal which it is receiving it is going to adjust the air fuel ratio back to the stechometric value.
- That means complete combustion.
- What happens when there is a lean mixture?
- When we have a lean mixture that means the there is less of fuel and more of oxygen.
- But something else happens here.
- The emissions now contain NOx gases like NO, NO2, etc.
- These NOx gases, if we want to get rid of them, because they also cause pollution, we know smog or acid rain is the result of these NOx emissions.
- But we are providing an oxidizing atmosphere because we have a lean mixture.
- There's a lot of oxygen and hence we find that NOPS emissions will increase if we have a lean mixture.
- Whereas if it is a rich mixture, that means the amount of oxygen is not sufficient to cause the combustion.

Example or application from the lecture

- Let us see further what does it do.
- We are just making sure that there is complete combustion and therefore only benign gases are let out into the atmosphere or less harmful gases are given out into the atmosphere.

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EV0000174 · EV0000175 · EV0000176 · +1 source units in pack.json

Core explanation · 10:00–14:59

harmful gases which are going to be let out into the atmosphere and like carbon monoxide and then also some carbon particles will be there and some unburnt fuel.

Key details

- So we understand that neither should it be a lean mixture nor should it be a rich mixture, and therefore the engine needs to work at the stechometric value that is 14.7 is to one and the oxygen sensor is helping us to do maintain that stechometric value.
- The cell is a solid state electrochemical fuel cell.

- Just now we discussed about different types of fuel cells and in that one of them was the solid state fuel cell.
- Oxygen sensor is basically an electrochemical fuel cell which is going to use solid as the electrolyte and we also know that when we use solid as electrolyte then we need then the cell will operate at a high temperature.
- So it is going to operate only at higher because we are using solid as the electrolyte.
- The anode here is platinum.
- Since it's a fuel cell, we know that there is going to be an anode, there is going to be cathode, and we have the exhaust gas on one side and the reference air on the other side because we are going we the oxy sensor has to compare the exhaust gas with the reference air.
- Electrodes which are used here are platinum electrodes.
- And the electrolyte, as I mentioned just now, it's a solid state.
- Electrochemical fuel cell.
- Reference air and of course these electrodes which are platinum.
- two to point eight.
- What happens is we have oxygen at the cathode.
- That is the reason why we chose zirconium oxide doped with Y2O.
- Three because we know that when zirconium oxide is duped with Y2O three, it creates oxide ion vacancies in the electrolyte.
- And since we have oxide ion which is formed at the ca at cathode and this oxide ion has to jump through the electrolyte and reach the anode so we have to provide this electrolyte which is a solid electrolyte zirconium oxide doped with y2o3.
- So these are the reactions which happen at the anode and cathode, and we can see that both the reactions are similar because it is like a concentration cell.
- Both sides at the anode as well as the cathode we have the oxygen which is present at the cathode we have reduction happening giving you O2 minus these O2 minus ions are moving towards the anode.

Example or application from the lecture

- Let us study the construction of the oxygen sensor.
- Let us study the working of this oxygen sensor.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000178 · EV0000179 · EV0000180 · +1 source units in pack.json

Core explanation · 15:00–19:59

And there they undergo oxidation, giving you O2 plus 4 electrolyte electrons.

Key details

- And the electrolyte is zirconium oxide doped with Y2O3.
- Solid electrolyte here because we need an electrolyte which is going to facilitate the movement of oxide ions.
- What would be the cell voltage?

- Now, since it is a concentration cell, we know that there will be no E naught, E naught value cancels values cancel for the anode and the cathode reactions, and therefore we have two.
- point three zero three r t over four f four is because of the number of electrons which are getting exchanged in these reactions so log of p1 by p2 now p1 is the partial pressure of oxygen in the reference air that is in the atmosphere and P2 is the partial pressure of oxygen in the exhaust.
- The reason for that is lead is going to poison the platinum catalyst.
- We know that we have used platinum at the anode and cathode.
- Poison the platinum catalyst, and that is going to reduce the efficiency of the oxygen sensor.
- When tetraethylaid is used because if the automobile is equipped with an oxygen sensor, this lead is going to poison the platinum and this will reduce the efficiency of the oxygen sensor.
- And we know that oxygen sensor is playing a very important role and therefore.
- Nowadays, unleaded gasoline is being used because most of the automobiles are equipped with oxygen sensor.
- For an oxygen sensor would be increased tailpipe emissions and increased fuel consumption and hesitation on acceleration.
- So in the next class we are going to talk about another set of energy storage devices and that is supercapacitors.

Example or application from the lecture

- The cell voltage can be given by the Nernst equation.
- Like for example methyl tertiary butyl ether.

These are edited study notes based on the cited lecture interval. Check uncertain speech recognition against the source transcript and recording.

EV0000182 · EV0000183 · EV0000184 · +1 source units in pack.json

A Symbol glossary

Symbol	Name	Meaning	Appears in

B Sources, coverage and gaps

Source	Path	Use
Gas sensor - O2 sensor in automobile exhaust stream, Biosensors – Blood glucose sensor (Part 1) transcript (JSON)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/037_Gas_sensor_-_O2_sensor_in_automobile_exhaust_stream_Biosensors_Blood_glucose_sensor_Part_1__58334f2a4af3fdff.transcript.json	Full JSON sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Gas sensor - O2 sensor in automobile exhaust stream, Biosensors – Blood glucose sensor (Part 1) transcript (Markdown)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/037_Gas_sensor_-_O2_sensor_in_automobile_exhaust_stream_Biosensors_Blood_glucose_sensor_Part_1__58334f2a4af3fdff.transcript.md	Full Markdown sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.
Gas sensor - O2 sensor in automobile exhaust stream, Biosensors – Blood glucose sensor (Part 1) transcript (plain text)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/037_Gas_sensor_-_O2_sensor_in_automobile_exhaust_stream_Biosensors_Blood_glucose_sensor_Part_1__58334f2a4af3fdff.transcript.txt	Full plain text sidecar. Normalized text similarity to canonical JSON: 0.9973; only the JSON segmentation supplies the displayed timestamps.
Gas sensor - O2 sensor in automobile exhaust stream, Biosensors – Blood glucose sensor (Part 1) transcript (SRT)	https://huggingface.co/buckets/alvin3y10/data/tree/pesuc/PESU_Materials/Sem-1/UE26CY151A_Engineering_Chemistry/Unit_03/AV_Summaries/037_Gas_sensor_-_O2_sensor_in_automobile_exhaust_stream_Biosensors_Blood_glucose_sensor_Part_1__58334f2a4af3fdff.srt	Full SRT sidecar. Normalized text similarity to canonical JSON: 1.0000; only the JSON segmentation supplies the displayed timestamps.

Historical question-source excerpts are shown only when a page has a strong text-based unit match. They are source references, not new solutions or exam predictions.

Known gaps, conflicts and corrections

1. 1 rendered pages in 034_Hydrogen_production_and_storage_Hydrogen_production_methods_Gasification_steam_reforming_photolysis_electrolys__bc4024e4ded7a197.slides.md have no readable `.slides.md` text; their original page images are included.
2. This is a visibly labelled source-preservation draft. Human page-level semantic and visual review is not complete; no unsupported subject summary or formula correction is claimed.
3. The 19.26 GiB MP4 files remain hosted on Hugging Face; the full video path/hash and transcript coverage are indexed. The full audio/video was not watched end-to-end.
4. Formula candidates are machine-extracted text with source locators; use the adjacent original slide page and the formula index before treating a candidate as an authoritative equation.
5. The study PDF contains edited, topic-organized lecture notes and textual slide-page takeaways. Full-page slide images remain in the source/visual index datasets, not embedded as the lesson. Human content, ASR, formula, and page-level visual review remains pending.

Coverage

Evidence is summarized by source to keep the reading copy legible. Full source-unit IDs, locators and decisions remain in pack.json.

Source	Unit counts	Example decisions
SRC00038	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0060. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00039	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0060. Sidecar variants compare...
SRC00040	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0060. Sidecar variants compare...

Source	Unit counts	Example decisions
SRC00041	4 covered · 0 excluded · 0 gaps	t000000-000299: Timestamp interval t000000-000299; canonical segment IDs s0001 to s0020. Sidecar variants compare... t000300-000599: Timestamp interval t000300-000599; canonical segment IDs s0021 to s0039. Sidecar variants compare... t000600-000899: Timestamp interval t000600-000899; canonical segment IDs s0040 to s0060. Sidecar variants compare...

C Revision and retrieval practice